12TH CHEMISTRY · CHAPTER 30 · FULL CHAPTER NOTES

Spectroscopy-2 NMR

Topic-wise short and long questions with answers from the complete chapter

INTRODUCTION

Q.1 [Short] What is NMR spectroscopy used for?

Nuclear Magnetic Resonance (NMR) spectroscopy is a powerful technique used extensively in chemistry, the polymer industry, the study of proteins and natural products, environment monitoring and the medical field. In chemistry it is particularly used to identify organic compounds and determine their structures. Proton (^{1}H) NMR and carbon-13 (^{13}C) NMR are the two most commonly used methods.

30.1 PRINCIPLE OF PROTON NMR

Q.2 [Long] Explain the principle of proton (^{1}H) NMR.

NMR spectroscopy is based on the magnetic properties (**nuclear spin**) of nuclei. Atoms with odd mass numbers, such as ^{1}H and ^{13}C, usually show NMR signals because their nuclei behave like tiny magnets.

1. In the absence of a magnetic field, all the nuclei have random orientation and the same energy.
2. In a strong magnetic field, the nuclei align either with or against the field, giving two spin states: a low energy state and a high energy state.
3. On irradiation with radio waves, nuclei in the low energy state absorb energy and move to the higher energy state (**spin flipping**).
4. When they return to their original state they emit energy, which is detected and converted into an NMR spectrum.

Q.3 [Short] Why is ^{12}C NMR inactive?

Atoms with even atomic and mass numbers have equal numbers of protons and neutrons in their nuclei. Their magnetic fields cancel each other, so such nuclei have no net magnetic field, e.g. ^{12}C. They are NMR inactive.

Q.4 [Short] What is meant by molecular environment? Explain with methanol.

The exact energy required to flip a nucleus depends on its position in the compound and the surrounding atoms, together called the **molecular environment**. So NMR can identify hydrogen atoms lying in different environments within one molecule. In methanol (CH_3OH), the three methyl hydrogens and the one hydroxyl hydrogen lie in different environments, so they absorb different frequencies of radio waves. Nuclei in different environments can also be flipped with radio waves of the same frequency if the strength of the magnetic field is varied.

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30.2 SOLVENTS FOR NMR SPECTROSCOPY

Q.5 [Long] Why are deuterated solvents such as $CDCl_3$ needed for a proton NMR spectrum?

The sample is dissolved in excess solvent. Ordinary solvents contain hydrogen atoms which give strong signals and interfere with the spectrum of the sample; e.g. the single proton of $CHCl_3$ would give a signal intense enough to mask the weaker signals of the sample. Therefore **deuterated solvents** such as $CDCl_3$ (deuterated chloroform) or CD_3OD (deuterated methanol) are used, in which H is replaced by its isotope deuterium (D). Deuterium needs a different radio frequency for flipping, so it neither absorbs the radio frequency given to the sample nor shows a peak in the ^1H NMR spectrum. The solvent is “silent” and only the signals of the sample are seen.

30.3 CHEMICAL ENVIRONMENT AND CHEMICAL SHIFT

Q.6 [Short] Define chemical shift. In which unit is it measured?

The position of a peak in a ^1H NMR spectrum relative to the TMS peak is called the **chemical shift (δ)**; it is measured in parts per million (ppm). It indicates the electronic environment of a proton: protons in different environments have different δ values, while those in the same environment have the same δ value. For example, in methanol the methyl protons appear at 3.35 ppm and the O–H proton at 4.80 ppm. The ^1H NMR range is 0–12 ppm.

Q.7 [Long] Explain the use of TMS as the standard for chemical shift measurement.

Tetramethylsilane, $Si(CH_3)_4$ (TMS), is the standard reference; its signal is assigned 0.0 ppm and all other signals are measured relative to it. It is suitable because:

1. All 12 of its protons have the same chemical environment and produce a single sharp peak.
2. Its protons appear at the extreme right of the spectrum (due to the low electronegativity of silicon) and do not overlap with most organic proton signals.
3. It is chemically inert.
4. It is volatile, which makes it easy to remove after use.

Q.8 [Short] What are chemically equivalent protons? Give examples.

All the protons in chemically identical environments within a molecule are **chemically equivalent** and show the same chemical shift. All the protons in TMS, methane, ethane, benzene, cyclohexane and propanone are equivalent, so each of these compounds gives a single peak. Molecules with chemically different sets of protons give more than one peak: propane and methanol have two environments; 1-chloropropane and ethanol have three.

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Q.9 [Short] How does electron density affect the chemical shift of a proton?

The chemical shift depends on the electron density around the proton. Protons surrounded by high electron density (alkyl groups) have lower chemical shift values, whereas protons with low electron density around them (near electronegative atoms such as F, O, N, Cl) appear at higher chemical shift values. In methanol, the methyl protons appear at 3.35 ppm and the O–H proton, next to electronegative oxygen, at 4.80 ppm.

Q.10 [Long] Give the chemical shift ranges of common types of protons.

Environment of proton	Examples	δ (ppm)
Alkyl (alkanes)	$-CH_3$, $-CH_2-$, >CH–	0.5 – 1.5
Alkyl next to C=O (ketone/ester)	CH_3–C=O, $-CH_2$–C=O	2.0 – 2.6
Alkyl next to aromatic ring	CH_3–Ar, $-CH_2$–Ar	2.3 – 3.0
Alkyl next to electronegative atom (halogenoalkanes, alcohols)	CH_3–O, $-CH_2$–O, CH_3–Cl, $-CH_2$–Cl	3.0 – 4.5
Vinylic (alkenes)	>C=CHR	4.5 – 6.0
Aromatic (attached to aromatic ring)	Ar–H	6.5 – 9.0
Aldehyde	RCHO	9.0 – 10.0
Alcohol	ROH	0.5 – 6.0
Phenol	Ar–OH	4.5 – 7.0
Carboxylic acid	RCOOH	9.0 – 13.0
Alkyl amine	R–NH–	1.0 – 5.0
Aryl amine	Ar–NH_2	3.0 – 6.0
Amide	RCONHR	5.0 – 12.0

Quick Check 30.1

Q.11 [Short] a) Explain why the 12 protons in TMS produce only a single peak.

In TMS, $Si(CH_3)_4$, the four methyl groups are attached to the same silicon atom in a symmetrical way, so all 12 protons have the same chemical environment (they are chemically equivalent). Equivalent protons have the same chemical shift, so only one sharp peak appears.

Q.12 [Short] **b) Identify how many chemical environments are there for protons in the following molecules. Also predict the expected chemical shift of the protons in each environment. i) 1,1,2,2-dichloroethane ii) propane iii) propan-2-ol iv) butan-2-one**

"1,1,2,2-dichloroethane" is not a correct name (four locants need "tetrachloro"). It is solved as 1,1,2,2-tetrachloroethane, $CHCl_2$–$CHCl_2$; 1,2-dichloroethane, CH_2Cl–CH_2Cl, gives the same answer (one environment).

Molecule	Environments	Protons and expected δ (ppm), Table 30.1
i) $CHCl_2$–$CHCl_2$	1	H on carbon bearing Cl: 3.0 – 4.5 (or higher, as two Cl atoms are attached)
ii) CH_3–CH_2–CH_3	2	–CH_3 (6H): 0.5 – 1.5; –CH_2– (2H): 0.5 – 1.5
iii) CH_3–CH(OH)–CH_3	3	–CH_3 (6H): 0.5 – 1.5; >CH–O (1H): 3.0 – 4.5; O–H (1H): 0.5 – 6.0
iv) CH_3–CO–CH_2–CH_3	3	CH_3–C=O (3H): 2.0 – 2.6; –CH_2–C=O (2H): 2.0 – 2.6; –CH_3 (3H): 0.5 – 1.5

Q.13 [Short] **c) What proton environment does a signal at δ 7.5 ppm likely represent?**

An aromatic proton (Ar–H, attached to an aromatic ring), whose range is 6.5 – 9.0 ppm.

30.4 INTERPRETATION OF PROTON NMR SPECTRA

Q.14 [Short] **What information does a low-resolution ^{1}H NMR spectrum give?**

A low-resolution ^{1}H NMR spectrum gives three pieces of information:

1. **Number of peaks** = number of different proton environments in the molecule.
2. **Position of each peak (chemical shift δ)** = type of environment, found by comparing with standard values (Table 30.1).
3. **Area under each peak** = relative number of protons in that environment.

Q.15 [Long] **Interpret the low-resolution ^{1}H NMR spectrum of ethanol.**

The low-resolution spectrum of ethanol (CH_3CH_2OH) has three peaks, so there are three proton environments: CH_3, CH_2 and OH. The peak at 5.4 ppm is the O–H proton (low electron density), the peak at 3.7 ppm is –CH_2– next to oxygen and the peak at 1.2 ppm is –CH_3. The peak areas are in the ratio 1 : 2 : 3, matching the number of protons in each environment; their sum gives 6 hydrogen atoms in the molecule.

Quick Check 30.2

Q.16 [Short] **a) What does the number of signals in a ^{1}H NMR spectrum represent?**

The number of different chemical environments of protons present in the molecule.

Q.17 [Short] b) Predict the number of peaks that will appear in the ^{1}H NMR spectra of the following molecules? (i) 1,1,2,2-dichloroethane (ii) butane (iii) propan-2-ol (iv) pentan-3-one (v) benzene

Molecule	Peaks	Reason
(i) $CHCl_2$–$CHCl_2$	1	both protons are equivalent
(ii) $CH_3CH_2CH_2CH_3$	2	two equivalent –CH_3 and two equivalent –CH_2– groups
(iii) $(CH_3)_2CHOH$	3	two equivalent –CH_3, the CH and the OH
(iv) $CH_3CH_2COCH_2CH_3$	2	two equivalent –CH_3 and two equivalent –CH_2– groups
(v) C_6H_6	1	all six protons are equivalent

Q.18 [Short] c) Predict the number of peaks and their chemical shifts in the ^{1}H NMR spectrum of bromoethane (CH_3-CH_2-Br).

Two peaks: –CH_3 protons at about 0.5 – 1.5 ppm and –CH_2–Br protons at about 3.0 – 4.5 ppm (next to the electronegative bromine).

Q.19 [Short] d) How many signals will be observed in the ^{1}H NMR spectrum of 2-chlorobutane. Draw its structure and identify the hydrogen atoms which will show the highest value of chemical shift.

Structure: CH_3–CHCl–CH_2–CH_3. It has four proton environments (CH_3 next to CHCl, the CHCl proton, the CH_2 and the end CH_3), so **four signals**. The hydrogen on the carbon bearing chlorine (–CHCl–) shows the highest chemical shift (3.0 – 4.5 ppm).

Q.20 [Short] e) What does the peak area in ^{1}H NMR represent?

The area under each peak is proportional to the number of protons giving that peak, so it shows the relative number of protons in each environment.

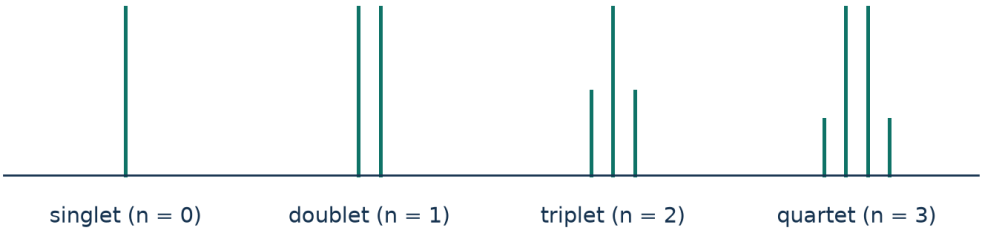
Q.21 [Long] What is splitting of peaks? State the n+1 rule and the common splitting patterns.

In a high-resolution spectrum each peak is often a cluster of sub-peaks. This **splitting** is caused by protons on adjacent (neighbouring) carbon atoms and shows how many of them there are.

Number of sub-peaks = n + 1 (n = number of protons on the adjacent carbon atoms)

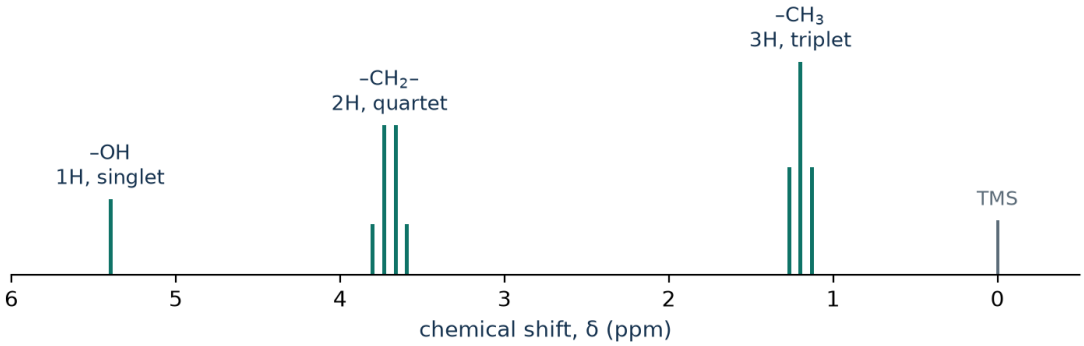
Protons on adjacent carbon (n)	Sub-peaks (n+1)	Splitting pattern	Abbreviation
0	1	Singlet	s
1	2	Doublet	d
2	3	Triplet	t

Protons on adjacent carbon (n)	Sub-peaks (n+1)	Splitting pattern	Abbreviation
3	4	Quartet	q



Q.22 [Short] Explain the splitting in the high-resolution spectrum of ethanol.

In the high-resolution spectrum of ethanol, the –CH_3 peak is a **triplet** because the adjacent carbon has 2 protons (2 + 1 = 3). The –CH_2– peak is a **quartet** because the adjacent carbon has 3 protons (3 + 1 = 4). The –OH signal is a **singlet**: it is not split because this hydrogen is bonded to oxygen, not carbon.



Sketch of the high-resolution ¹H NMR spectrum of ethanol

Q.23 [Short] In which cases is splitting not observed?

(1) –OH and –NH protons usually do not show splitting and appear as singlets. (2) Chemically equivalent protons do not split each other: in ethane (CH_3–CH_3) each methyl group has 3 protons on the adjacent carbon, yet the spectrum is a single unsplit peak because all six hydrogen atoms are equivalent.

Quick Check 30.3

Q.24 [Short] a) A signal in the ^{1}H NMR spectrum is a quartet. How many adjacent, equivalent protons ('n') are there?

$$n = \text{number of sub-peaks} - 1 = 4 - 1 = 3$$

Q.25 [Short] b) Calculate the number of sub-peaks for each signal in the high-resolution spectra of the following compounds? i) butan-2-ol ii) 1-chloropropane iii) propanone iv) 2,2-dimethylpropane v) methanol

Compound	Signal	n	Sub-peaks (n+1)
i) CH_3–CH(OH)–CH_2–CH_3	CH_3 (on C-1)	1	2 (doublet)
	CH (on C-2)	3 + 2 = 5	6
	CH_2	1 + 3 = 4	5
	CH_3 (on C-4)	2	3 (triplet)
	OH	–	1 (singlet, not split)
ii) CH_3–CH_2–CH_2Cl	CH_3	2	3 (triplet)
	middle CH_2	3 + 2 = 5	6
	CH_2Cl	2	3 (triplet)
iii) CH_3COCH_3	CH_3 (all six H equivalent)	0	1 (singlet)
iv) $(CH_3)_4C$	CH_3 (all twelve H equivalent)	0	1 (singlet)
v) CH_3OH	CH_3	0	1 (singlet)
	OH	–	1 (singlet)

Q.26 [Long] Describe the identification of O–H and N–H protons by proton exchange using D_2O.

The hydrogen atoms of O–H (alcohols, carboxylic acids) and N–H (amines, amides) are easily exchangeable, so their singlet appears over a wide range of chemical shift and is hard to identify. The **deuterium exchange** ("D_2O shake") is used:

1. The ^1H NMR spectrum of the compound is recorded; it shows all the H atoms including the O–H or N–H proton.
2. A few drops of D_2O are added to the NMR sample tube and the solution is shaken well.
3. The spectrum is recorded again.
4. The peak of the O–H or N–H proton disappears in the second spectrum. Comparing both spectra identifies that peak.

Protons attached to highly electronegative atoms like O or N are easily exchanged. When D_2O is added, these hydrogens are quickly exchanged with deuterium atoms:

$$R–O–H + D_2O \rightleftharpoons R–O–D + HOD$$

Deuterium needs a different energy for spin flipping than ^1H, so it shows no peak in the ^1H NMR spectrum; it is "invisible".

Quick Check 30.4

Q.27 [Short] **a) Why do O-H or N-H peaks disappear after a D_2O shake?**

The hydrogens of O–H and N–H are attached to highly electronegative atoms and are quickly exchanged with deuterium from D_2O (R–O–H becomes R–O–D). Deuterium needs a different energy for spin flipping, so it gives no peak in the ^1H NMR spectrum.

Q.28 [Short] **b) State the expected peak area and splitting patterns for ($-CH_2CH_2NH_2$).**

Peak areas: 2 : 2 : 2 ($-CH_2-$: $-CH_2-N$: $-NH_2$). The $-CH_2-$ next to nitrogen is a triplet (two protons on the adjacent carbon; N–H protons cause no splitting). The other $-CH_2-$ is split into a triplet by the CH_2-N protons (further splitting depends on the group attached to its other side). The $-NH_2$ protons give a singlet.

30.5 PREDICTION OF A PROTON NMR SPECTRUM

Q.29 [Short] **Give the steps for predicting the ^1H NMR spectrum of a molecule.**

1. **Step I:** Identify the different proton environments. Number of environments = number of peaks.
2. **Step II:** Estimate the chemical shift of each set of protons from Table 30.1. Without the table: protons near electronegative elements have higher δ; those on alkyl groups have lower δ.
3. **Step III:** Predict the splitting pattern of each peak by the n+1 rule.
4. **Step IV:** Assign integration (peak area) ratios according to the number of hydrogen atoms in each environment.

Q.30 [Short] **Predict the ^1H NMR spectrum of chloroethane. (Example 30.1)**

Protons	δ (ppm)	Splitting	Area
$-CH_3$	0.5 – 1.5	triplet (2 + 1 = 3)	3
$-CH_2-Cl$	3.0 – 4.5	quartet (3 + 1 = 4)	2

Chloroethane (CH_3-CH_2-Cl) has two proton environments, so two peaks with area ratio 3 : 2.

Quick Check 30.5

Q.31 [Short] **For the compound $(CH_3)_2CHOH$, predict the following: a) The number of peaks b) The type of proton and chemical shift c) The relative peak areas d) The splitting pattern**

a) Three peaks (three proton environments).

Type of proton	b) δ (ppm)	c) Area	d) Splitting
two $-CH_3$ groups	0.5 – 1.5	6	doublet (1 + 1 = 2)
>CH–O	3.0 – 4.5	1	7 sub-peaks (6 + 1; septet)

Type of proton	b) δ (ppm)	c) Area	d) Splitting
O–H	0.5 – 6.0	1	singlet

30.6 STRUCTURE FROM A PROTON NMR SPECTRUM

Q.32 [Short] How is the structure of an unknown compound deduced from its ^{1}H NMR spectrum?

1. Count the number of peaks: number of peaks = number of proton environments.
2. Use the chemical shift of each peak to identify functional groups (data table).
3. Use the peak area to find the number of hydrogen atoms in each environment.
4. Analyse the splitting pattern: n = number of sub-peaks − 1 gives the number of adjacent protons.
5. Combine this information to assemble the complete structure.

Q.33 [Long] The ^{1}H NMR spectrum of an ester shows peaks at 1.3 (3H), 2.2 (3H) and 3.9 ppm (2H). Deduce its structure. (Example 30.2)

δ (ppm)	Protons	Responsible group	Splitting	Neighbouring protons
1.3	3H	$–CH_3$ next to $–CH_2–$	triplet	3 − 1 = 2
2.2	3H	$–CH_3$ next to C=O	singlet	1 − 1 = 0
3.9	2H	$–CH_2–$ next to O	quartet	4 − 1 = 3

Combining: $CH_3–C(=O)–O–CH_2–CH_3$, i.e. **ethyl ethanoate**.

In the textbook table the $–CH_2–$ signal at 3.9 ppm is printed as "doublet, 2 − 1 = 1". A $–CH_2–$ next to $–CH_3$ has three neighbouring protons, so it is a quartet (as the 2H multiplet in the printed spectrum shows). Step 1 of the textbook also prints "No. of peaks = No of protons"; it should read "number of proton environments".

30.7 CARBON-13 NMR SPECTROSCOPY

Q.34 [Short] What is the basis of ^{13}C NMR? Give its reference, range and solvent.

The ^{12}C nucleus has an even number of protons and neutrons, has no magnetic properties and is NMR inactive. The ^{13}C nucleus has magnetic spin due to its odd number of neutrons, so ^{13}C NMR spectra can be recorded for carbon compounds. The chemical shift is again compared with TMS (δ = 0.0 ppm), but the range is 0–220 ppm. $CDCl_3$ is the usual solvent; it gives a small signal near 80 ppm.

Q.35 [Short] What is meant by the chemical environment of a carbon atom?

A carbon atom's environment is determined by the types of atoms directly attached to it. Carbon atoms in the same environment are chemically equivalent and show the same chemical shift; those in different environments appear at different positions. Both carbons of ethane are equivalent. In 2-methylpropane the three methyl carbons have the same environment, while the tertiary carbon has a different one.

Q.36 [Long] Give the chemical shift ranges of different carbon environments.

Environment of carbon atom	Examples	δ (ppm)
Alkyl	CH_3–, –CH_2–, >CH–	0 – 50
Carbon next to alkene or aryl ring	–C–C=C, –C–Ar	25 – 50
Carbon next to carbonyl or carboxyl	–C–COR, –C–COOR	30 – 65
Carbon next to halogen	–C–X	30 – 60
Carbon next to oxygen	–C–O	50 – 70
Alkene or aryl carbon	>C=C<, aromatic ring	110 – 160
Carboxyl carbon	R–COOH, –COOR	160 – 185
Carbonyl carbon	R–CHO, R–CO–R	190 – 220
Nitrile	R–C≡N	100 – 125

Q.37 [Short] How is the number of peaks in a ^{13}C NMR spectrum predicted? Give examples.

In a ^{13}C NMR spectrum each peak corresponds to one environment of carbon: **number of peaks = number of different carbon environments**. Ethane gives one peak. Propane (CH_3–CH_2–CH_3) gives two peaks: one for the two terminal –CH_3 carbons and one for the middle –CH_2– carbon. Ethanol (CH_3–CH_2–OH) gives two peaks: 18 ppm for the –CH_3 carbon and near 57 ppm for the –CH_2– carbon (next to oxygen).

Q.38 [Short] Give the differences between ^{1}H NMR and ^{13}C NMR spectra.

^{1}H NMR	^{13}C NMR
Peak area is proportional to the number of protons.	Relative peak area is usually not proportional to the number of carbon atoms.
Peaks show splitting (n+1 rule).	Peaks do not show splitting; they appear as vertical lines.
Chemical shift range 0–12 ppm.	Chemical shift range 0–220 ppm.
More sensitive.	Less sensitive.

Quick Check 30.6

Q.39 [Short] **a) How many signals do you expect to observe in the ^{13}C NMR spectrum of butane? Justify your answer.**

Two signals. In CH_3–CH_2–CH_2–CH_3 the two end –CH_3 carbons are equivalent and the two –CH_2– carbons are equivalent, so there are only two carbon environments.

Q.40 [Short] **b) Predict the ^{13}C NMR spectrum of butan-2-ol.**

CH_3–CH(OH)–CH_2–CH_3 has four different carbon environments, so four peaks (unsplit lines): the C–OH carbon at 50 – 70 ppm and the three alkyl carbons (CH_3 on C-1, CH_2, CH_3 on C-4) at different positions within 0 – 50 ppm.

Q.41 [Short] **c) Predict the chemical shift of carbon atoms in propanoic acid.**

Carbon in CH3–CH2–COOH	Environment	δ (ppm)
CH_3	alkyl	0 – 50
CH_2	next to carboxyl	30 – 65
COOH	carboxyl carbon	160 – 185

Q.42 [Long] **How is a ^{13}C NMR spectrum used to deduce the structure of a molecule?**

1. **Count the peaks:** the number of peaks gives the number of different carbon environments. If a molecule has 5 carbon atoms but only 3 peaks, there are 2 pairs of chemically equivalent carbons.
2. **Look at the position of each peak (δ):** it tells whether the carbon is in an alkane, alkene, alkyne, aromatic or carbonyl environment (Table 30.3).
3. **Work out the spectrum for possible structures:** draw the structures that fit the molecular formula and compare.

Example: a compound $C_4H_{10}O$ shows 3 peaks, so it has 3 carbon environments. Of the possible molecules butan-1-ol, butan-2-ol and 2-methoxypropane, the two alcohols would each show 4 peaks. The compound must be **2-methoxypropane, $(CH_3)_2CH$–O–CH_3**, which has 3 environments (two equivalent CH_3, the CH and the O–CH_3).

Quick Check 30.7

Q.43 [Short] **a) Draw the structures of butan-1-ol and butan-2-ol. Using the spectrum given in Figure 30.8, explain why this spectrum does not match with the structures of these two molecules.**

Butan-1-ol: CH_3–CH_2–CH_2–CH_2–OH. Butan-2-ol: CH_3–CH(OH)–CH_2–CH_3. In each molecule all four carbon atoms are in different environments, so each would give **four** peaks. The spectrum in Figure 30.8 has only **three** peaks, so it cannot belong to either; it fits 2-methoxypropane, $(CH_3)_2$CH–O–CH_3.

Q.44 [Short] **b) Which peaks (mention chemical shift) would you expect in the spectra of butan-1-ol and butan-2-ol?**

Butan-1-ol: one peak at 50 – 70 ppm (–CH_2–OH carbon) and three separate peaks within 0 – 50 ppm (two –CH_2– carbons and the –CH_3 carbon). **Butan-2-ol:** one peak at 50 – 70 ppm (>CH–OH carbon) and three separate peaks within 0 – 50 ppm (two –CH_3 carbons and the –CH_2– carbon).

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